Parkinson's and UPDRS Scores

Parkinson's disease is a brain disorder that makes people shake, stiffen up, have trouble keeping their balance and coordinating their movements (National Institutes of Health (NIH) n.d.). Typically, symptoms manifest progressively and worsen over time, where individuals may experience trouble walking and communicating. In determining the level of seriousness and progression of Parkinson's disease in patients, the Unified Parkinson's Disease Rating Scale (UPDRS) can be used (Macmillan Health Care Information 1987). Comprising of four sections, it was developed in 1987 to monitor how well the Parkinson's medications are working in order to alleviate symptoms. Instead of UPDRS scores being determined by expert physicians, machine learning techniques such as Multiple Linear Regression (MLR) models can be utilised to predict their scores based on a provided data set. Therefore, in this short study, a few MLR models using forward/backward validation set approach (VSA), k-fold cross-validation (CV), Ridge regression and Least Absolute Shrinkage and Selection Operator (Lasso) regression are compared amongst one another to obtain the model with the best UPDRS score predictions. Using models instead of manual labour in determining UPDRS scores may be beneficial in the sense that automated techniques can aid in early detection of the disease to prevent further progression, besides being much less time-consuming. It can also provide more consistent assessments since it reduces the variability between different evaluators, and is especially useful for large clinical settings where the number of patients is high.

Data Set

In order to compare the performance of different MLR models in predicting UPDRS scores, the "Parkinson Speech Dataset with Multiple Types of Sound Recordings Data Set" (Erdogdu Sakar et al. 2013) is utilised. The speech data set dating back to 2014, is a multivariate data set consisting of 26 sound recordings from 20 individuals with Parkinson's and 20 healthy individuals each. The sound recordings comprise of voice samples including sustained vowels, numbers, words and short sentences. Thus, it amounts to 1040 instances with 26 voice feature variables, subject id, UPDRS score and class information columns.

General Procedure

In creating MLR models, a model using all the predictors available in the Parkinson speech data set would have the following equation:

$$\hat{Y} = \widehat{\beta_0} + \widehat{\beta_1}X_1 + \widehat{\beta_2}X_2 + \cdots + \widehat{\beta_{26}}X_{26} \qquad \ldots (1)$$

In the analysis, there are 26 predictors consisting of the 26 voice feature variables, and the response variable is the UPDRS score. Before carrying out the analysis, the subject id and class information columns are dropped as they are irrelevant to the problem of predicting UPDRS scores and are more suited for classification problems. The 26 predictor variables are also renamed for personal convenience. The 80:20 rule is used for the training and testing sets of each method utilised. The Ridge regression and Lasso methods specifically, require the predictor variables to be scaled by dividing by their standard deviation before splitting since they work directly with the parameter coefficient sizes. This is to enable the variables to be treated equally and thus improve the interpretability of coefficients. The training and testing sets for all the methods in this study are the same since a $random\_state = 1$ is utilised.

It is then important for us to address that if at least one predictor is significant in predicting the response, is every predictor or only a subset significant? Therefore, the few MLR models used in this short study are first explained one by one before displaying the final comparison results in the subsequent sections. All the MLR methods in this study make their variable selection and comparisons between models using the following statistics:

- Adjusted R^2: R^2 is the proportion of variation in the outcome that is explained by the predictor variables. By accounting for the quantity of predictors and sample size, the adjusted R^2 offers a more precise assessment of the predictive ability of a regression model.
- Akaike Information Criterion (AIC): Calculated from the number of independent variables used to build the model and the maximum likelihood estimate of the model.
- Bayesian Information Criterion (BIC): The BIC is similar to the AIC but it applies a stronger penalty for model complexity in that it considers the number of observations in the formula.

- Residual Sum of Squares (RSS): Calculates the portion of a data set's variance that a regression model cannot account for. It gauges the error term's level of variance.
- Cross-validation Error (CV Error): The average error discovered throughout the cross-validation process, where it is a gauge of how well a model performs when faced with new untrained data.
- Mean Squared Error (MSE): Evaluates the average squared difference between observed and predicted values. It is a commonly used metric in evaluating the performance of a regression model as it provides a quantitative assessment of the extent to which the model fits the data. Commonly, software regression implementations try to obtain parameter values that minimise the MSE.

The particular statistics used for each method in variable selection will be explained in their respective subsections. However, in comparing between methods to choose the best MLR model, MSE is the main metric utilised as it enables direct comparison of different regression models by selecting the model with the lowest MSE. Only the best model chosen will be discussed in more detail.

Basic MLR Model

First off, we fit the basic MLR model, where all the coefficients are estimated using the least squares technique. All the 26 predictor variables are incorporated at first, leading to an adjusted R^2 value of 0.105, meaning that approximately 10.5% of the variation in the UPDRS score is explained by the predictor variables in the model. This means that our basic MLR model captures only a small proportion of the total variability in the UPDRS score. For a predictor variable to be significant, the condition is $p - value < 0.05$. It turns out that the whole basic MLR model is significant because the $prob(F - statistic) = 1.68 \times 10^{-13} < 0.05$. However, only the *Jitter_local_abs*, *Shimmer_apq11* and *Mean_period* are significant and have a relationship with the UPDRS score. Thus, we then try to fit an MLR model using only these three predictor variables as follows:

$$\hat{Y} = \widehat{\beta_0} + \widehat{\beta_1}\left(Jitter_{local_{abs}}\right) + \widehat{\beta_2}\left(Shimmer_{apq11}\right) + \widehat{\beta_3}\left(Mean_{period}\right) \qquad \ldots (2)$$

Turns out that this model with only the significant predictor variables from earlier does an even worser job since it only explains 4.8% of the variation in the UPDRS score. This could be due to the predictors having correlations with other predictors or the multicollinearity effect, which can impact the model's overall performance. The BIC however does decrease when using only three predictors. This indicates that the model's fit has worsened but its complexity has also decreased slightly. Since there are too many predictor variables to manually fit MLR models one by one, other types of fitting besides least squares such as stepwise selection processes can be utilised. Next, forward and backward validation set approach is carried out to find a better model.

Forward and Backward Validation Set Approach

Since the best subset selection is nearly impossible due to long computing times, forward selection is a simpler method of adding the best predictors to a current model. Here, we begin with an empty model and select the best predictor to add to it based on the statistic measure chosen. In backward selection, the opposite is done where we begin with a full model and remove variables one by one until the statistic measure is maximised. For our analysis, a forward and backward VSA is implemented, where the data set must first be split into a training and test set (80:20 rule). For the VSA to give correct estimates of the test error, we must only use the training observations for all parts of fitting the model. Thus, only the training data can be used to figure out which model of a given size is the best. For each model size, the best set of variables are determined as well as a returned test error. In choosing the appropriate number of predictors, the RSS value is computed on the test set. It was found that for the forward selection, the best model with the lowest RSS value of 43017.90 has 11 predictors. On the other hand, the best model for the backward selection has a RSS value of 43237.44 with 12 predictors.

K-fold Cross-validation with 5/6/7/8/9/10 Folds: Forward Selection

The cross-validation (CV) approach has k samples in the test set and uses all other $n-k$ samples to build a model. If there are $n = 100$ samples and $k = 10$, each fold will have 10 samples. Thus, 90 samples will be in the training set, whereas 10 will be in the test set with every fold rotating to be in the test set. In assessing the models, CV MSE is simply calculated by taking the average of the MSE for each fold. Technically, this evaluation is quite robust as the model is based on almost all of the data, where the splits are not random because each fold will only be excluded once. It is common practice to use between 5 to 10 folds as it balances the trade-off between bias and variance. The number of folds can be adjusted and experimented within this frame to get the appropriate number of folds for a particular data set. For our case, we carry out CV for forward selection using 5, 6, 7, 8, 9 and 10 folds. Thus, forward selection needs to be performed within each of the k training sets. The model with the lowest CV error among models of different sizes is chosen as the best model for each number of folds.

Number of Folds	**5**	**6**	**7**	**8**	**9**	**10**
Number of Predictors with Lowest CV Error score	15	11	15	10	13	12

Shrinkage Methods: Ridge Regression and Lasso Regression

Shrinkage methods will fit a model containing all the predictors in a data set, in our case all 26 predictors. The coefficient estimates are regularised causing them to decline towards zero, to reduce overfitting and multicollinearity. Doing so, there is no assurance that the fit will be improved even if the variance lessens. In Ridge regression, a penalty term ($\lambda \sum_{j=1}^{26} \beta_j^2$) is added to the traditional least squares function. A larger λ value will have stronger regularisation effects since the predictors will head to zero. Coefficient estimates are shrunk to nearly zero, which could reduce their importance. If the number of predictors in a data set were to be large, a Ridge regression model would be hard to interpret since all predictors are taken into account in the model. Lasso can solve this issue by introducing a penalty term that uses the absolute value of coefficient estimates instead ($\lambda \sum_{j=1}^{26} |\beta_j|$). By bringing some of the coefficients to exactly zero, Lasso accomplishes variable selection. As a result, only the most crucial predictor variables are kept in the model. In our analysis, CV techniques are utilised to find the optimal alpha value for both Ridge and Lasso. This is done using the cross-validated Ridge regression function, *RidgeCV()* for Ridge regression and 10-fold cross-validation using *LassoCV()* for Lasso regression. The MSE scores for both models are listed in the next section.

Comparing MLR Models

	Basic MLR Model		Validation Set Approach		K-fold Cross Validation: Forward Selection						Shrinkage Methods	
	All 26 Predictors	3 Predictors	Forward (11 predictors)	Backward (12 predictors)	5-fold (15 predictors)	6-fold (11 predictors)	7-fold (15 predictors)	8-fold (10 predictors)	9-fold (13 predictors)	10-fold (12 predictors)	Ridge	Lasso
MSE	220.85	235.48	186.09	184.66	212.00	209.52	216.40	246.18	214.18	210.07	218.83	219.08

In choosing the best MLR models from a few fitted models, the MSE score of the test set is observed, where a lower score indicates a better performing model. We observe that the basic MLR models have the worst performance, followed by Lasso and Ridge regression models. The Ridge and Lasso models might have performed badly due to the presence of many irrelevant predictor variables, in which they might have not been filtered out effectively by the regularisation techniques. Out of the 6 k-fold CV models, the 6-fold CV model performed the best with an MSE of 209.52. However, out of all 12 models, the backward VSA model performed the best with an MSE of 184.66, just slightly better than the forward VSA model. The plot below portrays that the best backward VSA model has 12 predictors since it has the lowest RSS value.

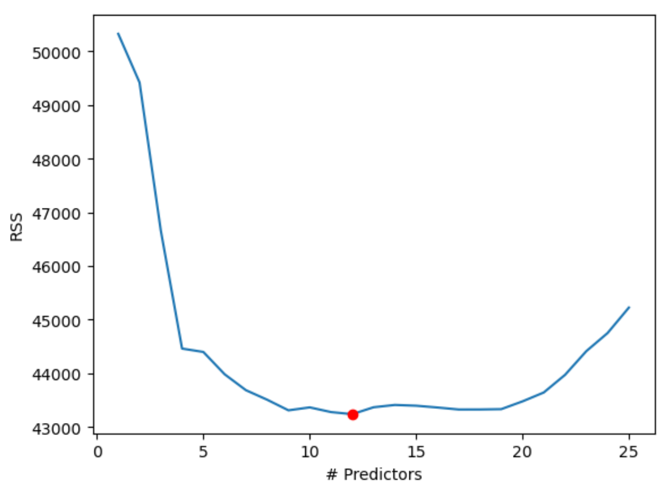

Figure 1 Plot of RSS values for backward VSA models of different predictor sizes

Best MLR Model for Parkinson Speech Data Set: Backward VSA

Since the backward VSA model with 12 predictors is selected as the best MLR model, we now fit the model on the full data set to obtain more accurate final coefficient estimates. This way, instead of just using instances from the training or testing set, all 1040 instances can be utilised for maximum information gain to potentially improve the model's performance. The output obtained from fitting the model on the whole data set is shown in the figure below.

```
                                 OLS Regression Results
=======================================================================================
Dep. Variable:            UPDRS_score   R-squared (uncentered):                   0.476
Model:                            OLS   Adj. R-squared (uncentered):              0.469
Method:                 Least Squares   F-statistic:                              77.70
Date:                Tue, 30 May 2023   Prob (F-statistic):                   6.09e-135
Time:                        23:11:23   Log-Likelihood:                         -4282.7
No. Observations:                1040   AIC:                                      8589.
Df Residuals:                    1028   BIC:                                      8649.
Df Model:                          12
Covariance Type:            nonrobust
=====================================================================================================
                                       coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------------------------
Jitter_local_abs                  1.603e+04   1.04e+04      1.547      0.122   -4300.385    3.64e+04
Jitter_ddp                           1.2036      0.365      3.297      0.001       0.487       1.920
Shimmer_local                        0.5828      0.243      2.400      0.017       0.106       1.059
Shimmer_apq3                        -0.6555      0.340     -1.930      0.054      -1.322       0.011
Shimmer_apq11                        0.4611      0.106      4.366      0.000       0.254       0.668
AC                                  30.1530      8.020      3.760      0.000      14.416      45.890
NTH                                -14.1690      7.802     -1.816      0.070     -29.479       1.141
Mean_pitch                          -0.0796      0.024     -3.334      0.001      -0.126      -0.033
Number_of_periods                    0.0108      0.004      3.068      0.002       0.004       0.018
Mean_period                      -2108.2080    649.477     -3.246      0.001   -3382.661    -833.755
Fraction_of_locally_unvoiced_frames  0.0624      0.030      2.069      0.039       0.003       0.122
Degree_of_voice_breaks              -0.1236      0.037     -3.333      0.001      -0.196      -0.051
==============================================================================
Omnibus:                      157.334   Durbin-Watson:                   0.191
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              230.435
Skew:                           1.102   Prob(JB):                     9.16e-51
Kurtosis:                       3.679   Cond. No.                     5.13e+06
==============================================================================
```

The final model, which does not contain a constant and thus is an uncentred model, is fitted on the entire data set and has the following MLR equation:

$$UPDRS_{score} = 16030(Jitter_{local_{abs}}) + 1.2036(Jitter_{ddp}) + 0.5828(Shimmer_{local}) - 0.6555(Shimmer_{apq3}) + 0.4611(Shimmer_{apq11}) + 30.1530(AC) - 14.1690(NTH) - 0.0796(Mean_{pitch}) + 0.0108\left(Number_{of_{periods}}\right) - 2108.2080(Mean_{period}) + 0.0624\left(Fraction_{of_{locally_{unvoiced_{frames}}}}\right) - 0.1236\left(Degree_{of_{voice_{breaks}}}\right) \quad \ldots (3)$$

Here, it means that the UPDRS score can be estimated based on the 12 predictor variables listed out. Each coefficient, while holding all other variables constant, indicates the change in the

UPDRS score for a one-unit increase in the relevant predictor variable. For example, a one unit increase in *NTH* would result in a 14.1690 decrease in the UPDRS score, assuming that all the other predictor variables stay constant.

The model is statistically significant with a $prob(F - statistic) = 6.09 \times 10^{-135} < 0.05$. Apparently, the adjusted R^2 score has increased by a big amount as compared to the full basic MLR model. Approximately 46.9% of the variation in the UPDRS scores can be explained by the model's predictor variables. Hence, our chosen model has a better fit and higher explanatory power. It is a good thing that we obtained this backward VSA model since our main focus is to achieve an MLR model with better predictive performance to predict UPDRS scores. Not to mention, since the full basic MLR model includes all 26 predictor variables, it is much more complex than our chosen model that only has 12 predictor variables. Noteworthily, with too many predictor variables, the model might capture random fluctuations or noise in the data set, which could lead to overfitting. This is when the model has poor generalisations on new and unseen data. A smaller model like the one chosen might be less prone to overfitting issues. A backward model is good since it begins with a complete model and gradually reduces variables to reveal redundant or insignificant predictors. <u>The final model mentioned above in Equation (3) can be used to predict UPDRS scores when provided with a new set of unseen Parkinson speech data.</u>

<u>References</u>

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